



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

SpaceX Falcon 9 First
Stage Landing Prediction
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Executive Summary

IBM's 90-launch Falcon 9 sample covers 2010–2020. Landing success is 60/90 (66.7%). Four classifiers were tested; the best held-out accuracy is 77.8% (14/18).

Introduction

Question: Which launch conditions are associated with first-stage landing, and can a model predict the outcome?

Scope: historical IBM course datasets, not current SpaceX operations. Class 1 means successful landing.



Section 1

Methodology

Methodology

Collect IBM course launch data and inspect missing values.

Summarize landing by site, orbit, payload and time.

Use SQLite for the supplied SQL exercise dataset.

Create site views and compare four classifiers.

Data Collection

Analysis uses four IBM course datasets: 90 launch records, 101 SQL exercise records, 56 dashboard records and 90 encoded feature rows. These samples differ; comparisons always state the source.

Data Collection – SpaceX API

Course snapshot of SpaceX API launch data → Falcon 9 filter → enrich rocket, payload, launchpad and core IDs → 90 launch records.

The course snapshot makes the analysis reproducible.

<https://github.com/delt-kz/ibm-spacex-capstone-2026>

API snapshot



Falcon 9 filter



Resolve IDs



Landing data

Data Collection - Scraping

The course scraping task extracts Falcon 9 launch rows from the Wikipedia launch history and standardizes dates, boosters and payloads. Here, the published IBM SQL CSV is used for reproducible query results; this report does not claim a fresh scrape.

<https://github.com/delt-kz/ibm-spacex-capstone-2026>

Launch tables



Parse rows



Normalize fields



SQL dataset

Data Wrangling

- The 90-row launch dataset has 26 missing LandingPad entries. Keep them as missing: no pad may mean no planned pad landing. Use the provided Class label, inspect categories, and one-hot encoded features for models. No outcome rows are removed.

EDA with Data Visualization

Charts compare landing outcome by launch site, orbit, flight number, payload mass and year. Group sizes are shown because rates from one or two launches are unstable.

EDA with SQL

SQLite queries reproduce the ten course tasks: unique sites, CCA launch rows, NASA payload, F9 v1.1 average payload, first ground landing, drone-ship cases, mission outcomes, maximum payload, 2015 failures and outcome ranking.

Build an Interactive Map with Folium

Plot the three distinct launch coordinates as markers. Color launch records by Class, then inspect site proximity. The course dataset gives coordinates but no rail, highway or coastline geometry; those distances are not inferred.

Build a Dashboard with Plotly Dash

Use the 56-row dashboard dataset for site-level pie charts and payload-versus-outcome exploration. Its rates differ from the 90-row API cohort, so the dashboard sample is labelled separately.

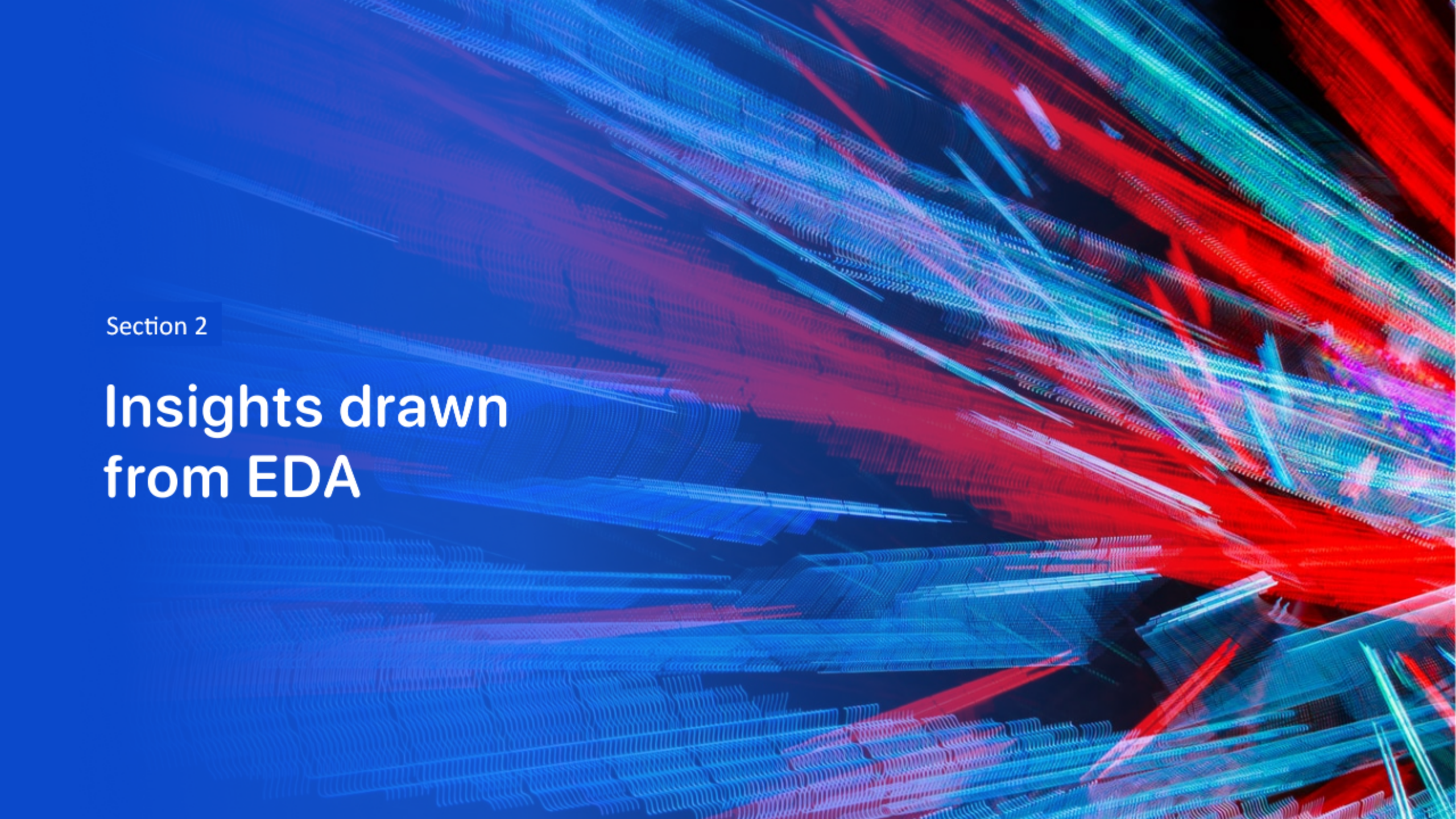
<https://github.com/delt-kz/ibm-spacex-capstone-2026>

Predictive Analysis (Classification)

Split 90 launches into 72 training and 18 test rows with stratification (seed 2). Scale features on training data only. Tune logistic regression, SVM, decision tree and KNN by five-fold cross-validation. Evaluate on the held-out test set.

Results

The 90-launch cohort contains 60 successful landings. Site success rates range from 60.0% to 77.3%. The best model reaches 77.8% held-out accuracy; the test set has only 18 launches.

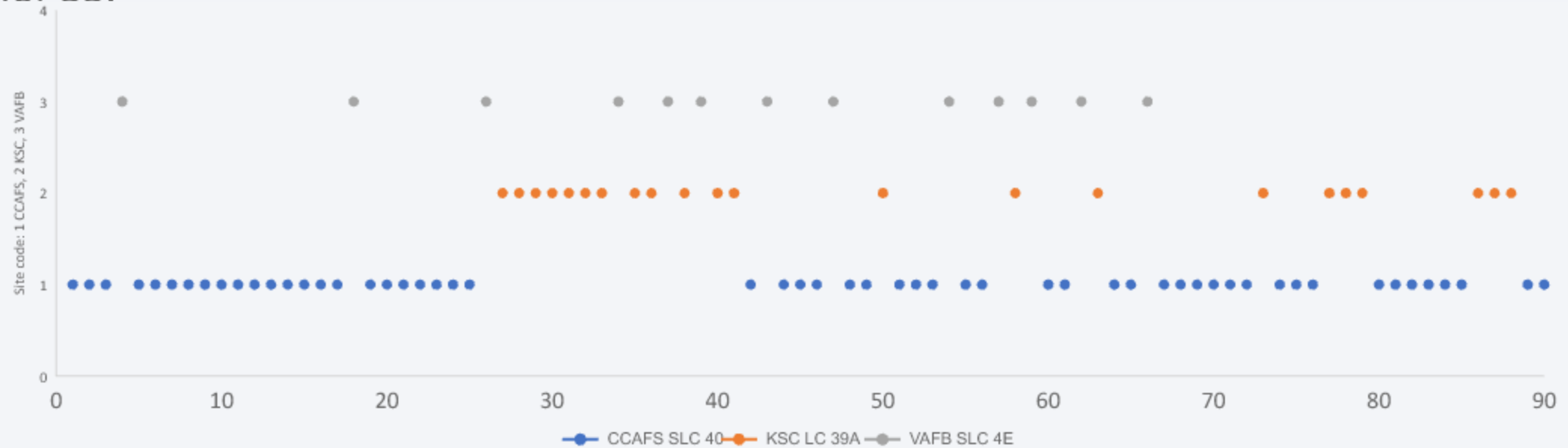
The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan, creating a sense of motion or data flow. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant.

Section 2

Insights drawn from EDA

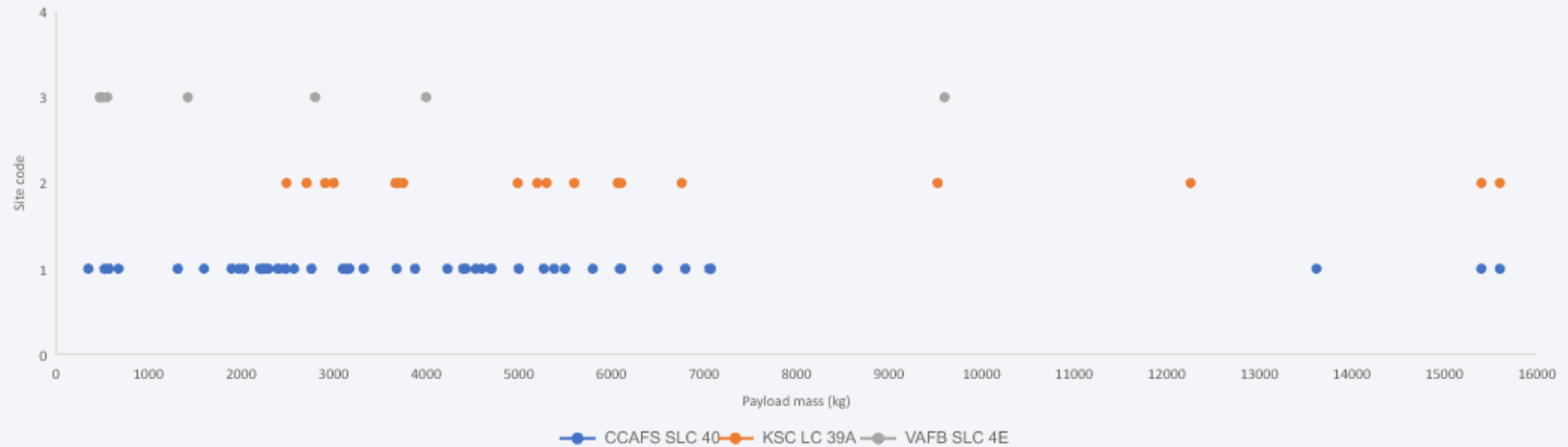
Flight Number vs. Launch Site

The 90 launches span three sites; CCAFS SLC 40 accounts for 55.



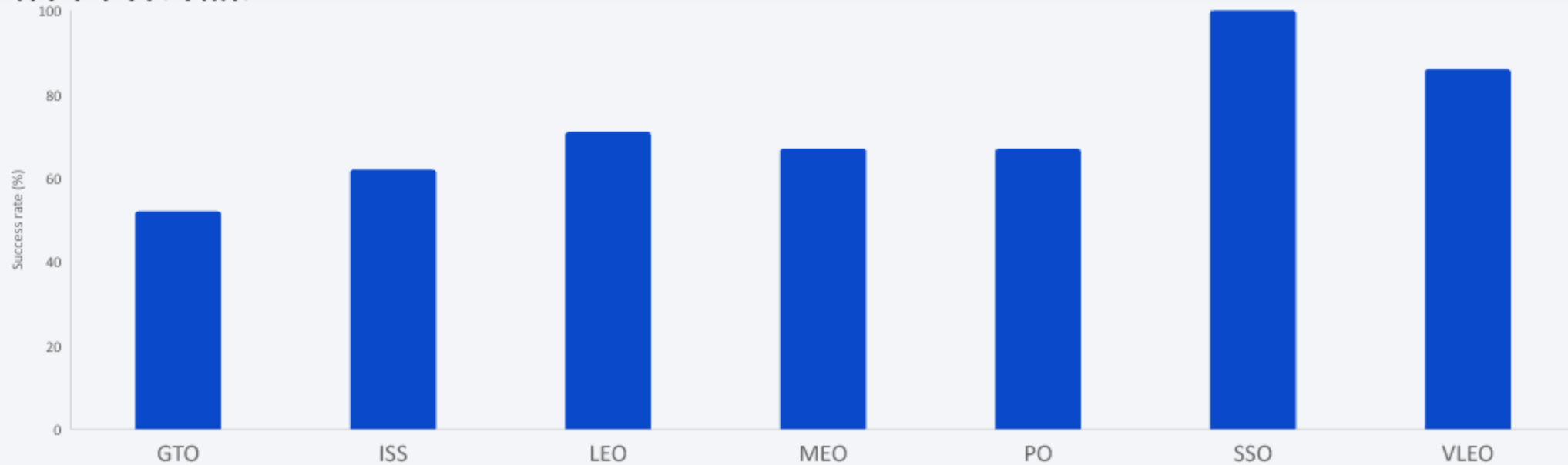
Payload vs. Launch Site

Payload mass ranges overlap across all three sites.



Success Rate vs. Orbit Type

VLEO has 12/14 successful landings (85.7%); GTO has 14/27 (51.9%).



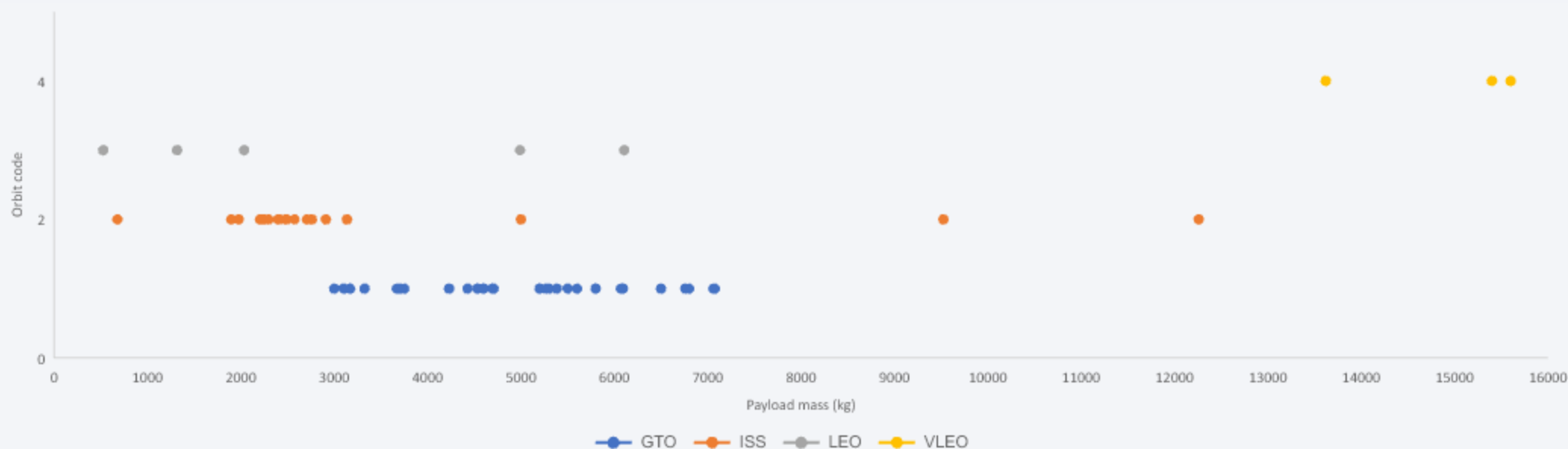
Flight Number vs. Orbit Type

Flight number approximates chronology; the orbit mix changes over time.



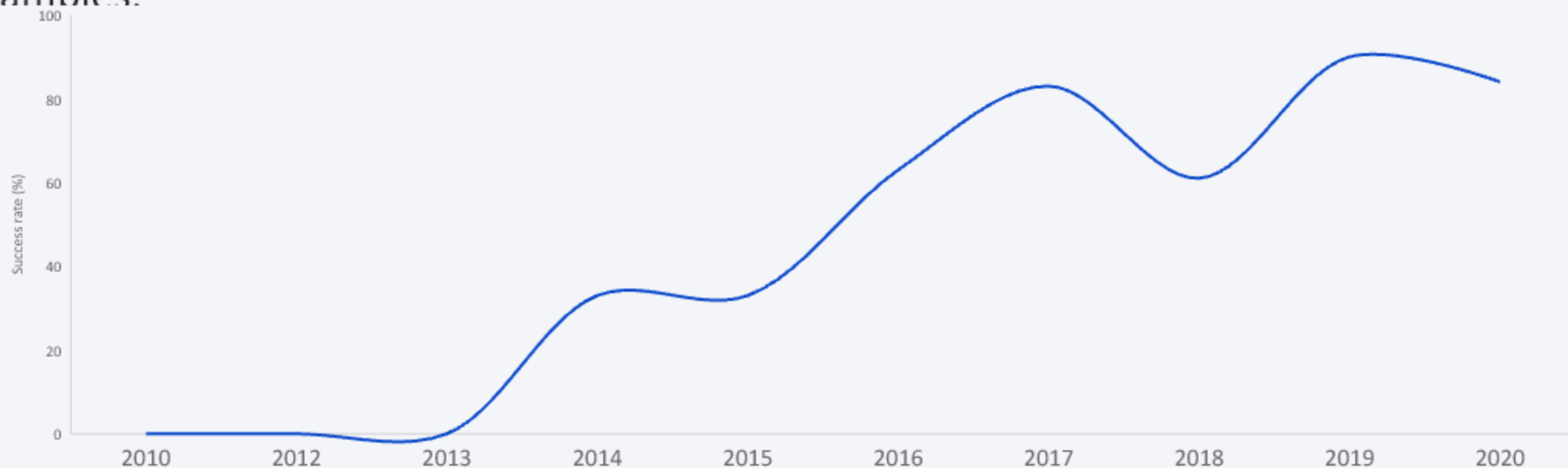
Payload vs. Orbit Type

Payload masses overlap across the four most common orbits.



Launch Success Yearly Trend

Yearly landing rates rise, but early years have small samples.



All Launch Site Names

Four distinct labels in the SQL sample:

CCAFS LC-40

CCAFS SLC-40

KSC LC-39A

VAFB SLC-4E

LC-40 and SLC-40 are separate labels in the source.

Launch Site Names Begin with 'CCA'

First five CCA rows:

2010-06-04 — CCAFS LC-40

2010-12-08 — CCAFS LC-40

2012-05-22 — CCAFS LC-40

2012-10-08 — CCAFS LC-40

2013-03-01 — CCAFS LC-40

Total Payload Mass

NASA-tagged customer rows carry 107 010 kg in total. Filter: Customer LIKE '%NASA%'.

Average Payload Mass by F9 v1.1

Average payload for F9 v1.1 variants: 2 534,7 kg. Filter: Booster_Version LIKE 'F9 v1.1%'.

First Successful Ground Landing Date

The first recorded successful ground-pad landing is 2015-12-22 in the SQL sample.

Successful Drone Ship Landing with Payload between 4000 and 6000

Drone-ship landings with 4,000–6,000 kg payload:

F9 FT B1021.2

F9 FT B1031.2

F9 FT B1022

F9 FT B1026

Total Number of Successful and Failure Mission Outcomes

Mission outcomes in 101 SQL rows:

Success: 98

Success (payload status unclear): 1

Success: 1

Failure (in flight): 1

The source has two differently spaced “Success” labels.

Boosters Carried Maximum Payload

Maximum payload is 15 600 kg. It occurs in 12 records across multiple F9 Block 5 boosters.

2015 Launch Records

Two 2015 drone-ship failures:

2015-01-10 — F9 v1.1 B1012 — CCAFS LC-40

2015-04-14 — F9 v1.1 B1015 — CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Landing outcomes through 2017-03-20:

No attempt: 10

Success (drone ship): 5

Failure (drone ship): 5

Success (ground pad): 3

Controlled (ocean): 3

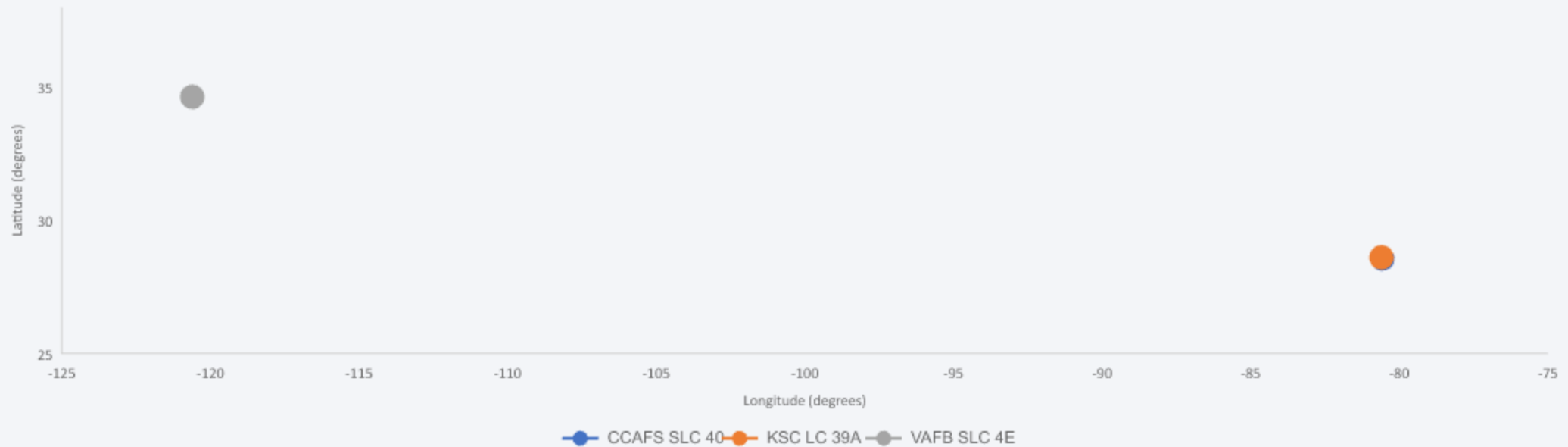
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a blue-tinted daytime view and a nighttime view showing city lights. The text is overlaid on the left side.

Section 3

Launch Sites Proximities Analysis

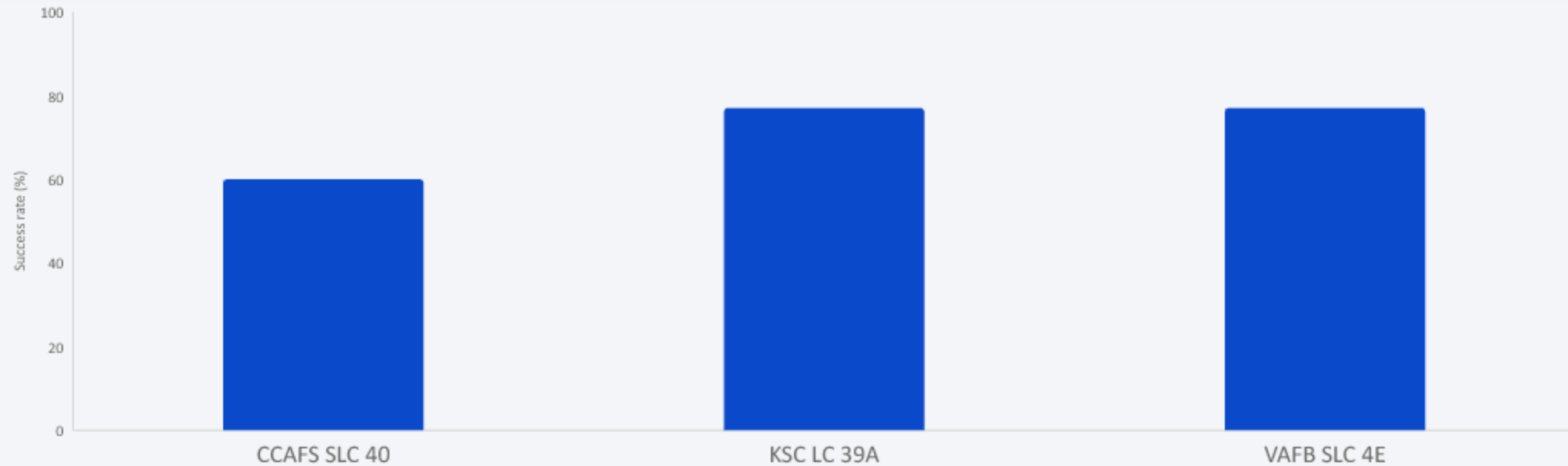
Falcon 9 launch site coordinates

Three distinct launch coordinates occur in the 90-row sample.



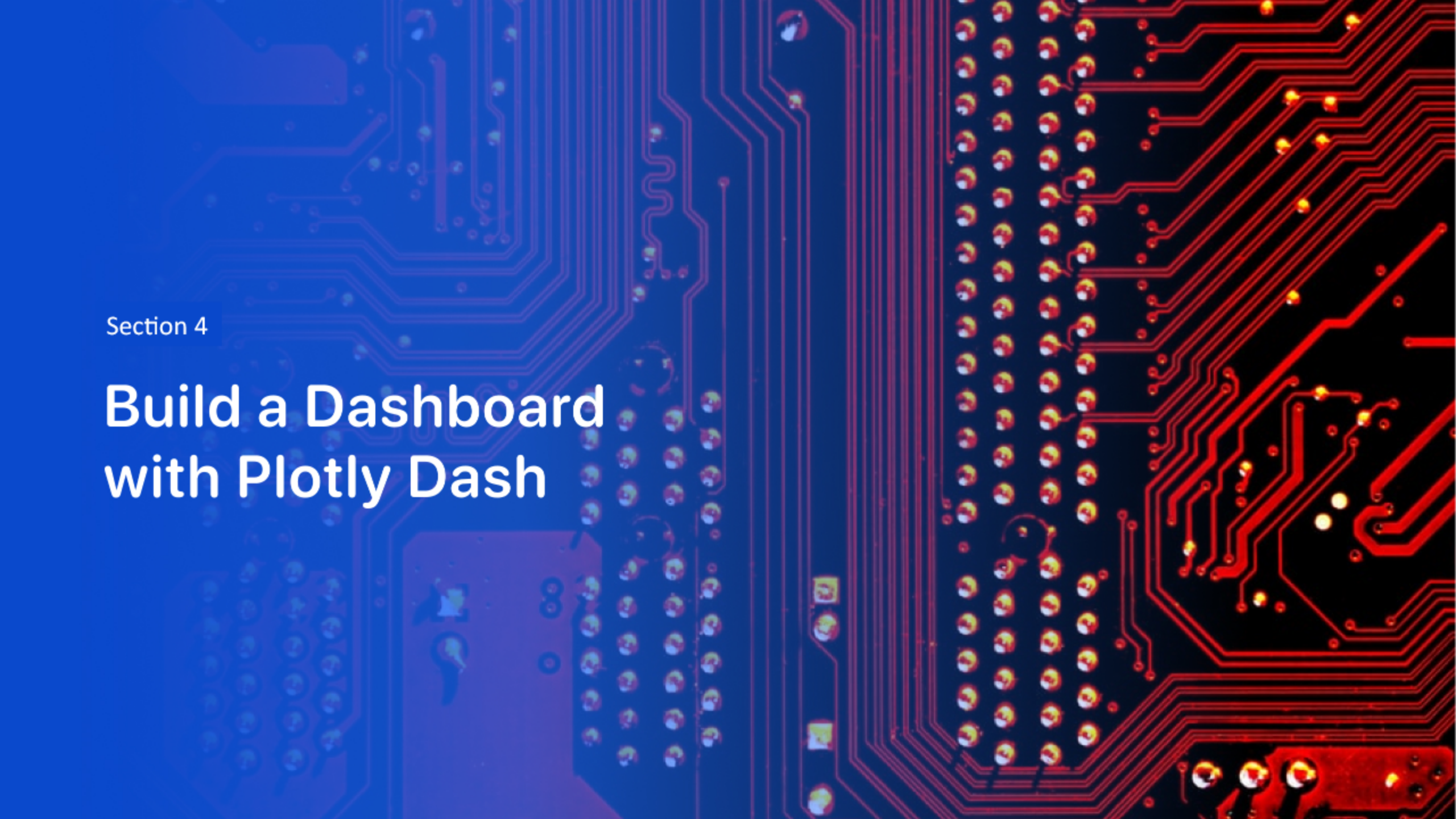
Landing outcome by launch site

Landing success ranges from 60.0% to 77.3% across sites.



Site proximity: data limitation

The IBM launch CSV has site latitude and longitude, but it does not include railways, highways or coastlines. This report leaves proximity distances unreported rather than inventing them. A geospatial base layer is needed for the course map task.

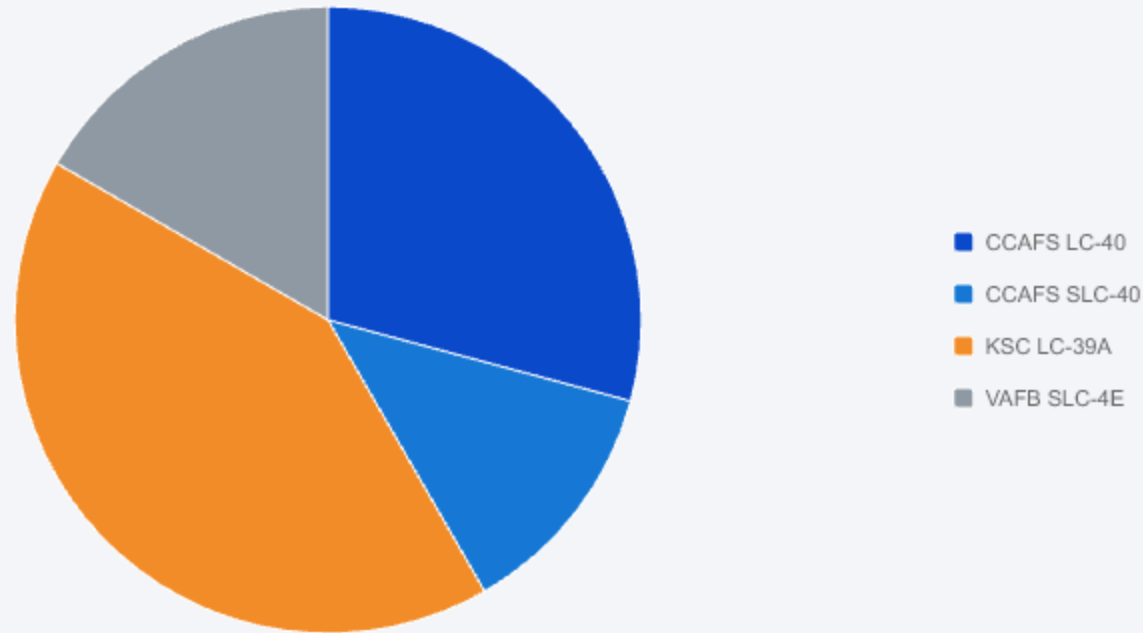


Section 4

Build a Dashboard with Plotly Dash

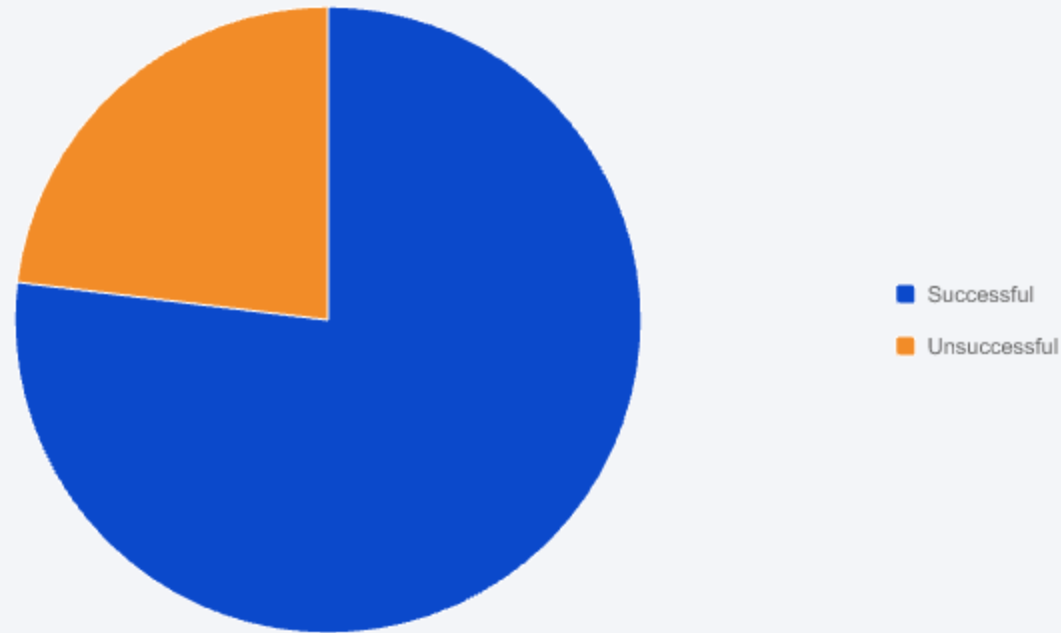
Dashboard sample: successful landings by site

The 56-row dashboard sample has 24 successful landings.



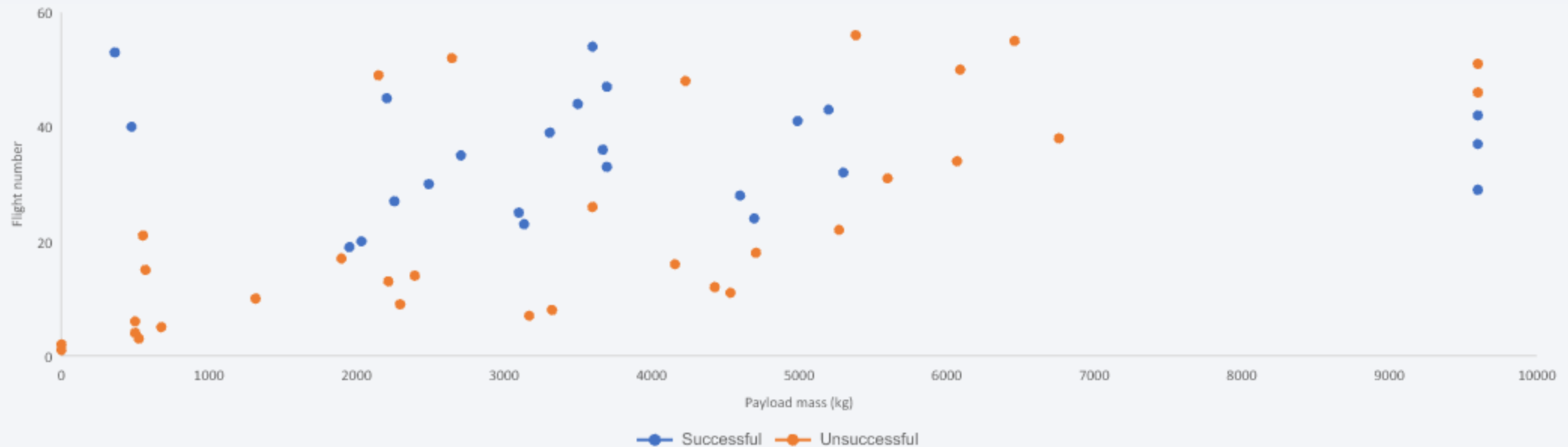
Dashboard sample: KSC success ratio

KSC LC-39A has the highest sample rate: 10/13 (76.9%).



Payload mass vs. launch outcome

The dashboard sample links payload mass and landing outcome.



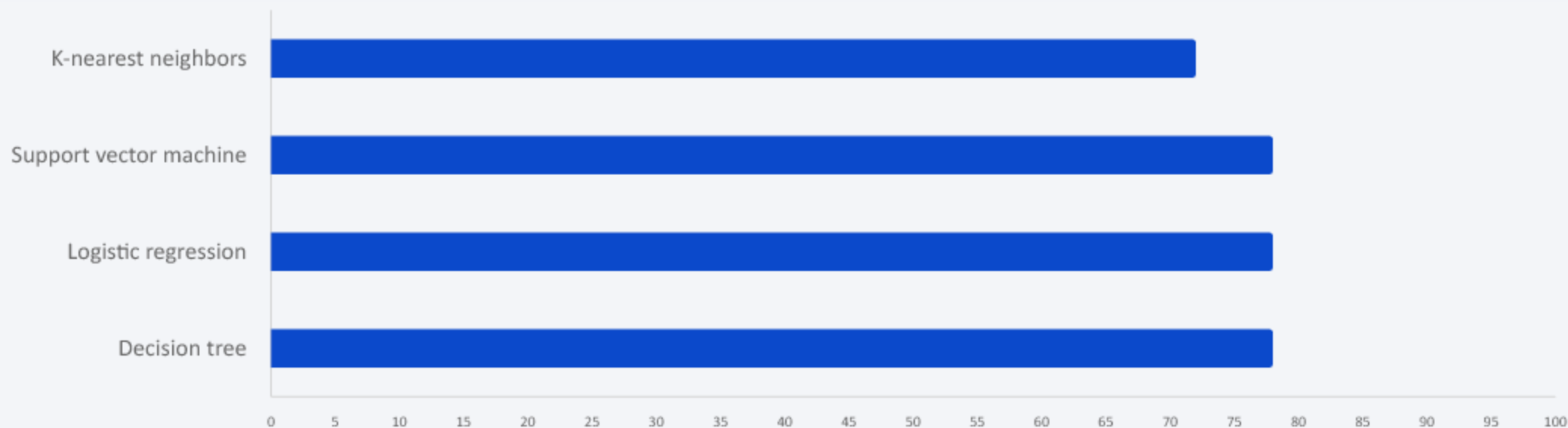


Section 5

Predictive Analysis (Classification)

Classification Accuracy

Three models tie at 14 correct predictions out of 18 test rows.



Confusion Matrix

Decision tree test confusion matrix (rows actual, columns predicted):

Actual 0: 5 predicted 0; 1 predicted 1.

Actual 1: 3 predicted 0; 9 predicted 1.

Accuracy: $14/18 = 77.8\%$.

Conclusions

Landing success rises with later launch years in this historical sample.

CCAFS SLC 40 has the lowest success rate among the three sites (60.0%).

The best held-out model score is 77.8% on only 18 launches.

Different IBM course datasets produce different rates; avoid merging them without reconciliation.

Appendix

Project: <https://github.com/delt-kz/ibm-spacex-capstone-2026>

IBM Skills Network datasets (links in speaker notes).

90 rows: EDA and models. 101 rows: SQL. 56 rows: dashboard.

AI-assisted analysis and report drafting; review before submission.

Thank you!

IBM Skills Network datasets · Historical sample through 2020 · AI-assisted draft

